



Federated CCTV Intelligence and Evidence Fabric

Hybrid of Models 1 + 2 + 3, with Model 4's central analytics on selected cameras. Model 4's statewide central recording is declined on arithmetic, not deferred.

30

GOVERNMENT
VERIFIED BASELINE

2

OWN FEED
DEMO

18

CONTROL SLOTS
REGISTRY ONLY · NO VIDEO

50

REGISTRY BASELINE
32 VIDEO SOURCES + 18 SLOTS

30 government + 2 own feeds with video; 18 control slots are registry rows without video.

Source: src/saakshya/command/domain.py::enforce_evaluation_50; var/live.db cameras (28 Sep 2026). Additional operator-onboarded cameras are retained. Screenshots are dated captures, not current totals.

What this deck is required to answer

PORTAL ASKS

- 1. Proposed solution model, with justification
- 2. Overview, objectives, and key innovations
- 3. High-level architecture and end-to-end workflow
- 4. AI video analytics — detection, recognition, events
- 5. Watchlist correlation and real-time alerts
- 6. Key technologies, frameworks, and tools
- 7. Scale, interoperability, security, deployment
- 8. Operational benefits and impact on policing

THIS DECK

Hybrid of Models 1 + 2 + 3, with selected-camera Model 4 analytics. Central VMS recording declined on 160 Gbps / 52 PB arithmetic.

Find → Trace → Verify → Act. Capability measured. Metadata moves. Video stays.

HLD-derived architecture and component interactions, then the pipeline from camera to search.

T0–T2 analytic stages. RT-DETRv2 + ByteTrack + voted ANPR. Live worker samples at a fixed interval; FRS is DESIGNED.

Match at ingest, same transaction as the sighting. Representative watchlist, labelled.

PyAV, PyTorch + ONNX Runtime, FastAPI, SQLite and PostgreSQL + PostGIS, vanilla ES. Licences pinned.

District edge + metadata centre. Four auth gates. Continues with the uplink down.

An index answers a plate that nobody had asked for yet. Honest grades, not a fake wall.

Sources: docs/HLD.md; docs/SCALE_MODEL.md. Screenshots retain their capture-time state; report measurements identify their own source.

“Participants will be provided with a designated vehicle registration number.”

By the time that number is handed over, the vehicle has already driven past the cameras.

There is no opportunity to go and look for it.

The only thing that can answer is an index of every plate already read — including the ones nobody asked for yet.

WHAT WE REHEARSED

Government grid

GJ11S7924 · cam06 only

Synthetic rendered test corpus

GJ18JX7786 · C-014 → C-021

Team stand-in; no real multi-camera evidence.

Gujarat's cameras are not one system

26 departments; the sandbox samples five: Home/Police, Health, GSRTC, Panchayat and Municipal bodies. Cameras differ in geometry, light and connectivity; plate-reading capability has to be measured.

1

Video cannot be centralised

80,000 cameras × 2 Mbps is 160 Gbps sustained. Thirty days is ~52 PB. MODELLED arithmetic; not a load test.

2

Capability is unknown and unequal

Capability is assessed from each camera's stream. Poor geometry and light constrain ANPR. UNKNOWN remains an explicit result.

3

Connectivity is unreliable

District-local processing can continue during uplink loss; queued metadata is replayed when the link returns.

Answer “where did this vehicle go?” without moving the video

ONBOARD	One registry for a heterogeneous estate. Codec, department and capability are data, not assumptions.
FIND	An index of every plate already read — including the ones nobody had asked for yet.
TRACE	Typed legs. Clocks ALLOWED or REFUSED. A looping camera is not a journey.
ACT	A watchlist hit is a row with a camera and evidence pointers. Video stays where it is.

KEY INNOVATIONS

Measure, never assume Three grades per camera per time band. UNKNOWN is a correct answer.	Metadata, not video 1,331.7 B/observation MEASURED; ~400 B MODELLED. The measured comparison is 19.3× smaller than video.
Match in one transaction Watchlist hit and sighting commit together, or neither does.	Graph before appearance Appearance ranking is not used as the lead. Graph and timebase checks come first.

Sources: var/reports/bandwidth.json; docs/SCALE_MODEL.md (~400 B model). Copilot tools are read-only; src/saakshya/copilot/tools.py.

Hybrid of Models 1 + 2 + 3, with selected-camera Model 4 analytics.

MODEL	ROLE / SOURCE PATH	IMPLEMENTATION / BOUNDARY
M1 — Registry & GIS (mandatory, kept)	Control plane, not a side deliverable. Identity, geometry, transport, health, measured capability.	Registry, GIS, governance and gap analysis. Cameras without coordinates remain listed; location provenance is retained.
M2 — Direct unified viewing + selective analytics	Direct integration with reachable cameras/NVRs over RTSP/ONVIF or departmental APIs. No federation middleware on this path.	Government streams and own-feed MediaMTX use direct access. Both register in Model 1. Wall policies are described separately.
M3 — VMS federation middleware	Departmental VMS → vendor API/SDK adapters → federation layer → unified platform. Both paths share the Model 1 registry.	Adapter contract and DEMO/TEST connectors built; no vendor SDK client or ONVIF discovery yet. Deployment is DESIGNED.
M4 — Central analytics (kept, selected cameras)	Selected streams through one controlled gateway into central analytics, events, watchlist, evidence, GIS.	Intelligence view. Detection, tracking, ANPR, alerts on own and selected feeds.
M4 — Statewide central recording (declined)	The transport, not the capability. Central ingest of all video refused on cost and on the Core Goal.	Not built, and not deferred. 80,000 × 2 Mbps ≈ 160 Gbps. 30-day retention ≈ 52 PB.

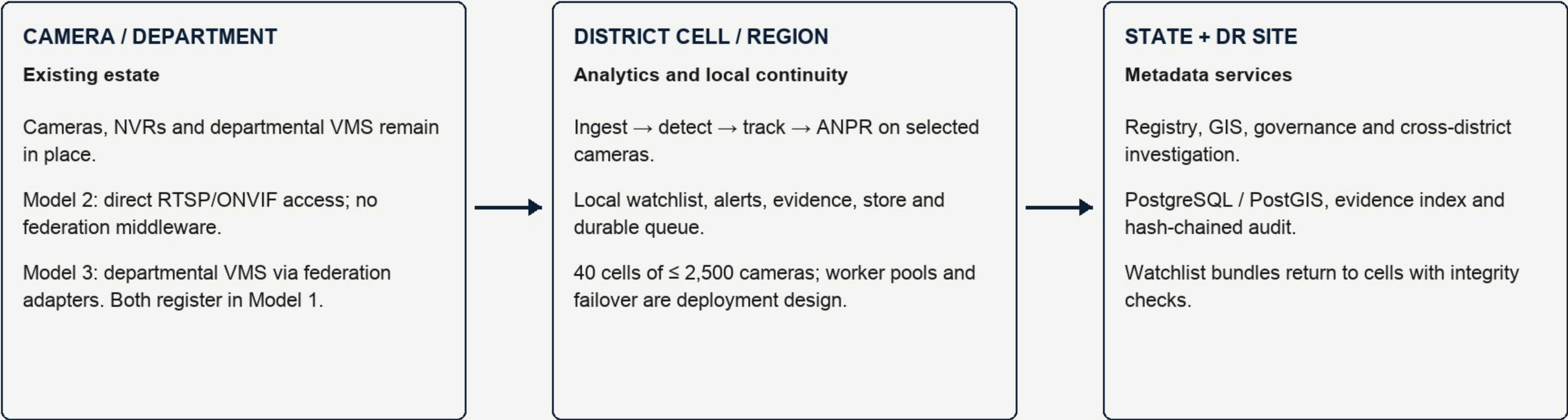
Sources: organiser FAQ Q12–Q23 (Model 2 direct; Model 3 middleware; hybrid allowed); docs/HLD.md §10; docs/SCALE_MODEL.md. Federation deployment is DESIGNED.

Why statewide central recording is declined

BASIS	RESULT	STATUS
80,000 cameras × 2 Mbps	160 Gbps sustained video ingest	MODELLED · HLD §1
160 Gbps × 30 days ÷ 8	≈52 PB of central video	MODELLED · HLD §1
Video demux from 3 cameras	1.363 Mbps	MEASURED · bandwidth.json
Serialised observation row	1,331.7 B	MEASURED · bandwidth.json
Peak stored event rate × row size	0.071 Mbps · 19.3× smaller	Report comparison · bandwidth.json
Compact observation assumption	~400 B per observation	MODELLED · SCALE_MODEL.md

Sources: docs/HLD.md §1; docs/SCALE_MODEL.md; var/reports/bandwidth.json. Selected-camera Model 4 analytics remain part of the hybrid; statewide central recording is declined.

Regional analytics; central metadata and governance



Video and metadata at regional ingest; metadata to the centre. Authorised viewing pulls video on demand.

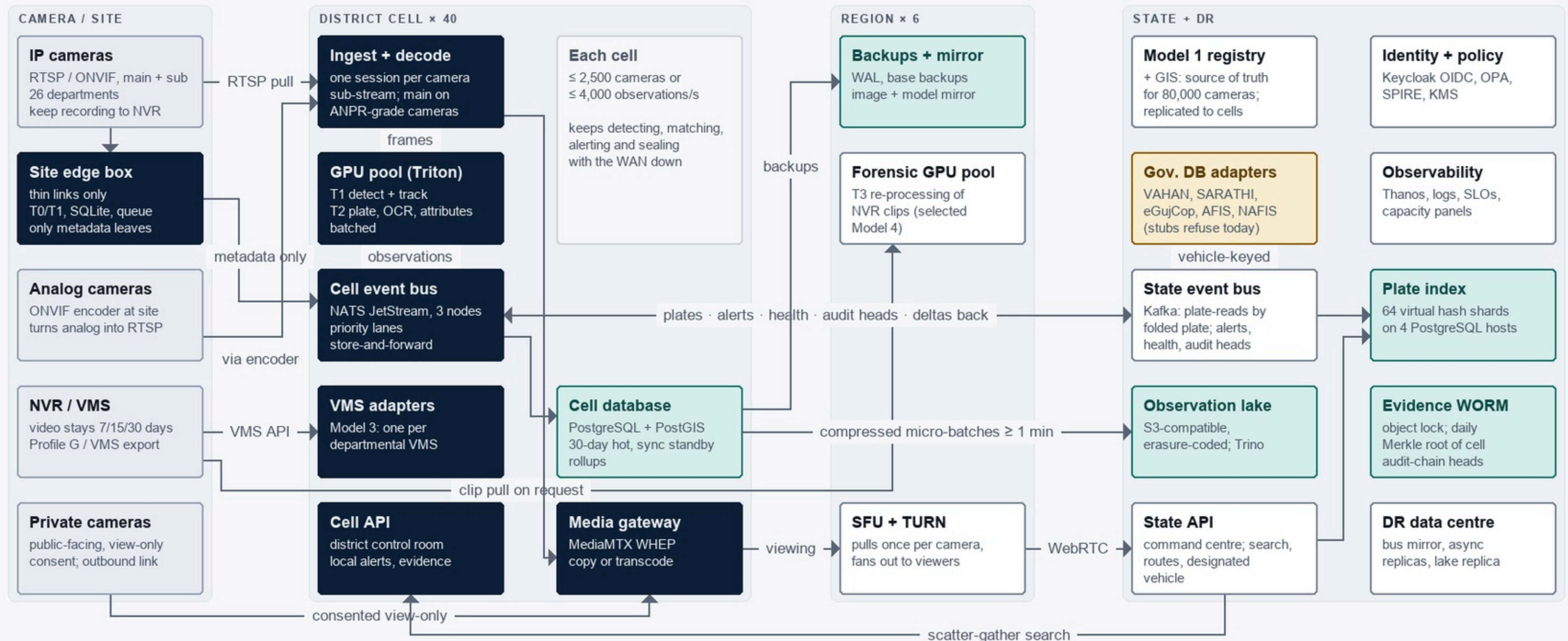
Models 1 + 2 + 3 form the foundation; selected-camera Model 4 analytics are included.

Statewide central recording is declined. Multi-node HA and DR require deployment validation.

Sources: organiser FAQ Q16–Q19, Q23; docs/HLD.md §3, §6, §10, §15, §21. DESIGNED topology; VERIFIED component contracts. Departmental retention stays in place.

Statewide target architecture

Video stays where it is recorded; metadata, events, requested clips and viewed streams move



DESIGNED (docs/STATEWIDE_ARCHITECTURE.md §2–§6; HLD §3, §6, §21). Registry and watchlist deltas, DR mirroring and evidence replication are in the boxes. Only inference compute grows in proportion to cameras analysed (MODELLED, reports/capacity_model.json). Nothing here ran at 80,000 cameras.

Ingest → analytics → edge → store → investigation

COMPONENT	INTERFACE / RESPONSIBILITY	BOUNDARY
Ingest and timebase	PTS-aware frames, reconnect/backoff and discontinuity checks feed camera pipelines.	A shared timeline requires reliable clocks.
Detection, tracking and ANPR	Vehicle/person boxes → track → plate crop → voted registration or no read.	Selected-camera inference at a fixed interval. Priority cadence and tier selection: tested in harnesses; live-worker wiring DESIGNED.
Local edge services	Sighting + watchlist match commit together; alerts, sealed evidence and durable queue remain local.	Uplink loss delays aggregation; local processing continues.
Store and metadata exchange	Idempotent replay into observation, evidence and audit stores. SQLite locally; PostgreSQL / PostGIS at scale.	Regional event bus and worker cluster are DESIGNED.
Investigation and access	Graph-first search, trajectory, GIS, incident workflow, trace report and read-only copilot.	Authentication, role, jurisdiction and purpose gates.

Source: docs/HLD.md §4–6, §10–11, §18, §20.6. Components are VERIFIED in the implementation; distributed deployment remains DESIGNED.

Test scenario, answered

BRIEF REQUIREMENT	WHERE TO LOOK	EVIDENCE AND BOUNDARY
~50-feed requirement / registry baseline	Cameras / registry · Live wall	30 government + 2 own feeds with video; 18 control slots are registry rows without video (capacity placeholders). Additional onboarded cameras are retained.
Centralised monitoring + AI	Overview · Live wall · Intelligence	Government RTSP / WebRTC and own feeds in one workspace; vehicle/person boxes and plate reads. Departmental systems remain in place.
Designated vehicle → route	Investigate · Trajectory legs	GJ18JX7786: C-014 → C-021. SYNTHETIC RENDERED TEST CORPUS — route-logic demonstration, not camera footage. Government stand-in GJ11S7924: cam06 only. No real multi-camera evidence is established.
Timestamped, location-wise history	Vehicle trace report · 06_* reports	Camera/location, timestamped reads, timed legs, sealed stills and row digest. Coverage gaps stay explicit; unreliable shared clocks are REFUSED.
Continuous watchlist cross-reference	Alerts · own/government films	Each ingested sighting is matched; hit and sighting commit together. Automated alerts show the read, priority and evidence; repeated reads form incidents.

Sources: src/saakshya/command/domain.py::enforce_evaluation_50; var/demo/SUBMIT/06_*trace_report.html. View names identify evidence; film chapter timestamps are not recorded here.

Compute, bandwidth and retention

RESOURCE	PLANNING BASIS — NOT A PROCUREMENT BENCHMARK
Cell compute	Per full cell: 2,500 cameras at 1 Hz, ANPR-grade 20% (ASSUMED) at 5 fps; S=10 ASSUMED over the 5.6 fps CPU sizing baseline. 81 GPUs + 2 spares in 21 four-GPU servers (12 if all at 1 Hz); 10 ingest/decode servers. HLD §20.2–20.3.
Hardware classes	Inference: 256 GB RAM per server. Decode: 64 cores / 256 GB. Database: primary + synchronous standby, 32 TB NVMe per copy. HLD §20.3; MODELLED, target GPU unmeasured.
Metadata bandwidth	1,331.7 B/observation MEASURED (bandwidth.json); batches compress 8.0× (measure_compression.json). At 3,000 observations/camera-hour (ASSUMED): 3.83 Mbps per full cell, 82 Mbps statewide — MODELLED, HLD §20.4.
Hot / warm / cold	MODELLED: 30-day hot metadata 14.4 TB per full cell (4.3 TB at the measured mean rate); lake 324 TB/year, 1.5 PB usable for 3 years plus a DR copy; evidence 100 TB write-once, assumed. HLD §20.5.
Low connectivity	Video remains at departmental NVRs. Optional site inference ships metadata only; durable queues replay after reconnect (a week is 194 GB per cell). Viewing and evidence retrieval add traffic. HLD §20.1 / §20.4.

Sources: docs/HLD.md §20.1–20.5; reports/capacity_model.json; var/reports/bandwidth.json. Retention and target hardware require departmental agreement.

80,000 cameras: only compute grows with cameras analysed

- Video stays with the department. Only metadata crosses the WAN: 82 Mbps statewide compressed at a pessimistic 3,000 observations per camera-hour (ASSUMED), against 160 Gbps for central video. MODELLED.
- A district cell holds at most 2,500 cameras or 4,000 observations/s, and runs detection, ANPR, alerts and evidence locally. Plates are partitioned by plate: a designated-vehicle route reads one shard. DESIGNED.
- Camera-driven non-compute resources keep $\geq 5\times$ headroom on throughput and $\geq 2.2\times$ on hot storage at 100% analysed; the 3-year lake (1.5 $\times$) is bought per retention year. MODELLED.
- Unit costs measured on this project: 1,331.7 B per observation, 8.0 $\times$ batch compression, 14,187 rows/s PostgreSQL, 1.3 ms PostGIS viewport on 80,000 cameras. GPU speed-up $S = 10$ is ASSUMED.

ANALYSED CAMERAS	GPUS INCL. SPARES	CELL WAN, MBPS	STATE BUS, MB/S	CELL DB ROWS/S	LAKE TB / YEAR
8,000	338	0.31	0.57	167	32.4
20,000	723	0.77	1.43	417	80.9
40,000	1,366	1.53	2.87	833	162
80,000	2,652	3.07	5.73	1,667	324

Growth at pessimistic rates, average cell of 80,000 / 40 cameras. 2,652 GPUs = 663 servers; a full cell sends 3.83 Mbps. Source: reports/capacity_model.json; HLD §21.

Three tiers, and what survives each failure

- Camera / site: nothing new at most sites; an edge box only where the site's link cannot carry its streams.
- District cell (40): GPU pool, event bus, PostgreSQL + PostGIS, media gateway, federation adapters. It keeps detecting, matching, alerting and sealing evidence with the WAN down.
- Region (6): viewing fan-out, backups, forensic GPUs. State + DR: registry and GIS, plate index, lake, identity.
- Model 1 everywhere; Model 2 direct; Model 3 for departments with a VMS; Model 4 on selected cameras only.

WHAT FAILS	WHAT KEEPS WORKING	WHAT DEGRADES
Cell WAN	Detection, local watchlist match, alerts, evidence sealing, district control room	Statewide search stale for that cell; catches up by priority lane
Region	Cells and state	Remote viewing moves to a neighbour region; backups queue
State event bus	Cells	Plate and alert lanes queue in the cell (7 days: 194 GB per cell, MODELLED)
State database	Cells; the bus keeps 7 days	Statewide routes pause, then replay
Whole state DC	Everything local	DR takes over; cells keep the last valid watchlist, fail-closed
GPU saturation	ALERT and HIGH_PRIORITY cadence first	NORMAL frames shed

Sources: docs/STATEWIDE_ARCHITECTURE.md §2, §8; HLD §3, §6, §15; tests/e2e/test_offline_mode.py. RPO/RTO targets ASSUMED; multi-node failover untested.

Every resource against its capacity: compute binds first

- Would-be bottlenecks found and designed out: uncompressed metadata on a 20 Mbps link (0.85×), every observation through the state bus (1.13×), a cell of highway cameras (0.91× on the database at half the measured write rate; cells now split at 4,000 observations/s).
- Inference compute binds first by construction: GPUs are bought to demand, S is measured on tendered hardware before purchase.
- Bus, gateway, TURN and object-store capacities are ASSUMED and are Phase 1 measurement gates (HLD §16).

RESOURCE	DRIVEN BY	DEMAND	CAPACITY	HEADROOM
Cell WAN, compressed	cameras	3.83 Mbps	20 Mbps	5.2×
Cell WAN, uncompressed (rejected)	cameras	23.5 Mbps	20 Mbps	0.85×
State bus, real-time lanes	cameras	5.73 MB/s	100 MB/s	17.4×
State bus, all observations (rejected)	cameras	88.8 MB/s	100 MB/s	1.13×
Cell database writes	cameras	2,083 rows/s	14,187 rows/s	6.8×
Cell hot store, 30 days	cameras	14.4 TB	32 TB	2.22×
State lake, 3 years	cameras	971 TB	1,500 TB	1.54×
Media gateway sessions / cell	viewers	200 sessions	450 sessions	2.25×
State API	users	5,000 req/s	12,331 req/s	2.47×
Watchlist bootstrap, one cell	list size	166 s	3,600 s	21.6×

Headroom = capacity ÷ demand. Storage is policy-elastic; viewers and users have their own admission control. Source: reports/capacity_model.json; HLD §21.

Indicative cost and avoided central-video spend

SCOPE	IMPLEMENTATION	ANNUAL OPERATIONS
PoC · approximately 50 cameras	INR 29–63 lakh	INR 18–50 lakh
Full district cell · 2,500 cameras	INR 8.9–20.4 crore	INR 1.4–3.7 crore
Statewide · 40 cells + 6 regions + state/DR	INR 308–707 crore	INR 51–134 crore
Avoided central video storage	INR 52–156 crore · disks only	52 PB; one copy, before DR / replication
Avoided central video transport	160 Gbps continuous ingest	INR 58–154 crore / year
Operational benefit	Searchable history, incident grouping, sealed evidence	Investigation time savings remain estimates; measure in the district pilot.

Source: docs/HLD.md §20.8–20.9; reports/capacity_model.json. MODELLED on ASSUMED S=10, 20% ANPR-grade at 5 fps and unit prices, before taxes; not quotes. VMS spend continues.

District cell pilot, regional rollout, metadata state tier

DESIGN / PHASE

TARGET AND VALIDATION

Cybersecurity	Segment camera, analytics, data and operator zones. TLS in transit; encrypted stores and backups; keys held apart. RBAC, jurisdiction and purpose gates; audit exported for review. HLD §18.
Cell continuity	Watchlist, evidence and queues continue locally during uplink loss. Cell metadata targets: RPO 0 in the cell, ≤5 min off it; RTO ≤60 s failover, ≤8 h rebuild. Assumed targets; restore/replay must be drilled. HLD §15.
State HA and DR	Synchronous standbys, event bus mirrored to DR, erasure-coded lake. Plate index and registry: RPO ≤1 min to DR, RTO ≤1 h at DR; lake reads ≤4 h. Multi-node failover remains untested. HLD §15.
Pilot → cell → region → state	Survey cameras, obtain catalogue and watchlist authority; each phase passes measured gates first (S on the tendered GPU, observation rate, compression ≥4×, WAN-cut drill). HLD §13–16.
Inputs from each department	Inventory, mount, codec, resolution and surveyed coordinates; NVR/VMS vendor, version and authorised API/stream access; network path/bandwidth; retention/evidence policy and watchlist authority. HLD §13.
Operations and acceptance	Health/readiness probes, metrics, structured logs and hash-chained audit exist. Deployment adds load balancing, log retention, alerts, camera assignment and failover. HLD §20.6.

Sources: docs/HLD.md §13–16, §18, §20.6; docs/STATEWIDE_ARCHITECTURE.md §8. RPO/RTO are ASSUMED targets, not measured recovery. Government integrations require authority.

A vehicle passes a camera. Only metadata moves.

- ① Registry already knows the camera, its position, and its adapter.
- ② Stream opened — timing from the video's own PTS, never from arrival.
- ③ Motion / vehicle / person detected and tracked. Plate located inside the vehicle box.
- ④ OCR on each sampled crop → per-track vote → one plate string, or no string.
- ⑤ Sighting written always. Watchlist match, if any, commits in the same transaction.
- ⑥ Operator sees the alert with its evidence pointers. Video stays where it is.
- ⑦ Later, a plate is typed. Graph-first retrieval returns what the cameras actually saw.

Two answers we will not invent

A face is presence, not identity. A shared timeline is REFUSED unless both cameras' clocks are sound.

UNKNOWN is a correct grade. A guessed colour is worse than no colour.

Detection, recognition, and event analytics

T0 · presence

Motion differencing per colour channel. A camera that cannot read a plate can still prove something passed.

T1 · detection + track

RT-DETRv2 vehicles and persons. ByteTrack reimplemented for real PTS and scene cuts. Person box is presence, not identity.

T2 · recognition

Plate found inside the vehicle box. OCR on sampled crops → per-track vote → one string, or no string. No face identification.

Events

Watchlist match, motion presence, long-stay duration, and restricted-zone entries. The government film shows the cam12 administrator DEMONSTRATION rule; intrusion accuracy is unmeasured.

FRS is DESIGNED and gated, not shipped: authorised gallery and legal basis, suitable camera grade, case/purpose and audit, human verification of candidates (HLD §11.2).

Sources: docs/HLD.md §4.2; runtime/inference_scheduler.py; runtime/budget.py. Tier selection and priority cadence are built and tested in harnesses; live-worker wiring is DESIGNED. Live sampling uses a fixed interval.

Every plate read is checked. A hit is a row, not a toast.

1	POST	Plate, category, authority and reason. An entry with no stated authority cannot be created.
2	SCOPE	Versioned, jurisdiction-scoped, with expiry. Labelled REPRESENTATIVE — not a live government record.
3	MATCH	Ingest compares every voted plate to the active local watchlist. Continuous while cameras are open.
4	COMMIT	Alert and sighting write in the same transaction, or neither does. There is no “alert without evidence”.
5	ALERT	OPEN, with decomposed confidence (plate × quality × category), recommended action, observation ids.
6	HOLD	Acknowledge → investigate → clear (with a reason). No delete. Every transition is in the hash-chained audit log.
7	EDGE	The district cell keeps matching if the uplink is down. Watchlist bundles fail closed on integrity.

VAHAN, SARATHI, eGujCop (CCTNS), AFIS and NAFIS: adapters refuse with SourceUnavailable. Departmental access is required (DESIGNED); no live database integration is claimed.

Sources: docs/HLD.md §10–11; src/saakshya/watchlist/government.py; watchlist/alerts.py. Local representative entries are separate from government database access.

What actually runs — licences pinned in code

LAYER	CHOICE	WHY THIS, NOT THE ALTERNATIVE
Decode / ingest	PyAV (FFmpeg)	Only practical option that exposes real PTS. cv2.VideoCapture does not.
Detection	RT-DETRv2-R18 · PyTorch on the GPU	Apache-2.0. Vehicles, persons, two-wheelers. Ultralytics refused (AGPL).
Tracking	ByteTrack, own implementation	Upstream is neither PTS-aware nor scene-cut-aware. Both matter here.
Recognition (ANPR)	YOLO-v9-t plate + Indian-trained OCR	Awiros-ANPR-OCR (Apache-2.0), ported to PyTorch. Per-track voting; recogniser evaluation is separate from the older films.
Store	SQLAlchemy · SQLite and PostgreSQL 18 + PostGIS	Both exercised; hash chains verified in the store migration report. ST_DWithin for cameras near a point.
API	FastAPI + Uvicorn	OpenAPI generated from routes. Four authorisation gates on every call.
Workspace	Vanilla ES modules, canvas map	No third-party asset. Runs with no internet route. Strict CSP is enforceable.
Local replica	MediaMTX	Mirrors the organiser sandbox. Own-feed demonstration store.

Sources: docs/HLD.md §5; var/reports/store_engines.json; var/reports/ocr_indian_eval.json. No frozen API or test counts.

District edge does the work. The centre holds the index.

Scalability

The 80,000 figure is a registry/GIS metadata load test with synthetic camera rows plus architecture sizing. Regional worker pools are DESIGNED; statewide live inference has not been tested.

Interoperability

RTSP/TCP, HLS and WHEP. Mixed h264/hevc, colour and IR. Adapters, not a replacement VMS. Government grid plus local MediaMTX. ONVIF-style metadata-first onboarding. Edge bundle export for disconnected nodes.

Security

Four gates: authentication, role, jurisdiction, purpose. ADMIN cannot search. AUDITOR sees cameras, not what they saw. Watchlist read is purpose-bound. Evidence and audit are hash-chained. No face identification. Stream URLs never returned to investigators.

Deployment

One codebase, three profiles (DEV_CPU, CLOUD_GPU, TARGET_GPU). Edge: ingest, analytics, local store, queue, watchlist, alerts, evidence. Centre: aggregation, cross-district search, GIS, audit. Continues with the uplink down. PostgreSQL + PostGIS store migration was measured; NATS at scale is designed; the films ran on SQLite + HTTP.

Sources: docs/HLD.md §10, §12, §18, §20.6; var/reports/final/model1/registry_80000.json. Vendor adapters are contracts; deployed multi-vendor certification remains a field task.

The shift gets an index, not another video wall

For the investigating officer

- Type a plate. See what cameras actually saw.
- Know which cameras cannot read a plate before wasting a shift.
- A watchlist hit names the camera and the observation.
- A lookalike is labelled. A looping pass is not a route.

For the organisation

- Uses cameras already installed. No 52 PB central store.
- Uplink loss leaves district-local detection running.
- Every intrusive query is purpose-bound and hash-chained.
- Districts keep matching when the wide-area link is down.

For policing and public safety

- A designated mark can be searched the moment it is handed over.
- Alerts fire where the camera already is — video does not have to move first.
- Honest UNSUITABLE stops false confidence on unreadable mounts.
- An audit trail answers who looked, at what, and why.

Source: docs/HLD.md §20.9. Operational benefits are expected; the avoided central video store is MODELLED. No statewide live-camera test is claimed.

Two explicit wall policies

POLICY / PLANE	BEHAVIOUR
CONTROL ROOM · Dense 6×5	One direct WHEP session per tile, up to 30; opens staggered 400 ms apart. A tile session is viewing, not deep inference.
OPTIMIZED VIEW · default	At most 12 sessions near the viewport; 600 px prefetch. Releases a session 15 s after its tile leaves the viewport.
Authenticated signalling	Browser WHEP uses SAAKSHYA’s authenticated signalling proxy. Sentinel credentials stay out of the browser. Selected AI streams use RTSP/TCP separately.
Shared sandbox availability	No fixed participant-facing concurrent RTSP limit. Availability varies with shared load: stagger, back off and isolate each camera. Report live counts only for their test window.

Sources: ui/app.js::applyMediaPolicy, TILE_WHEP_BUDGET, TILE_WHEP_STAGGER_MS, WHEP_PREFETCH_PX, WHEP_KEEPALIVE_MS; docs/SENTINEL_SANDBOX.md support clarification.

What it took to integrate the live grid

19 SEP · REPORTED TO SENTINEL

MEASURED in the retained 16 Sep UTC reports: 30/30 IDs yielded RTSP frames; 15 direct WHEP, 15 bridge candidates.

VideoToolbox reached 15 preview bridges in the scale test. The mixed wall held 19 browser-visible cameras for 60 s: 15 direct + 4 bridged; 0 NO_SIGNAL.

That wall classified 8 LIVE + 11 PREVIEW. Fan-in failures did not establish a bridge maximum or a sandbox quota.

SENTINEL · GUIDANCE

We asked about concurrency, long-lived sessions, the 30-camera pattern, a catalogue and connection pacing.

No fixed participant-facing session limit; availability varies with shared load. Open only needed streams; design independently of camera count.

Isolate cameras, stagger opens and back off. No separate /api/ingest catalogue. Fan-in variation is not a local-bridge limit.

VERIFIED · IMPLEMENTATION

Direct WHEP via authenticated signalling; credentials stay server-side. CONTROL ROOM / OPTIMIZED VIEW state the local session policy.

On-demand hub; 4 deep-inference slots, prioritised by measured capability. Viewing is separate from AI coverage.

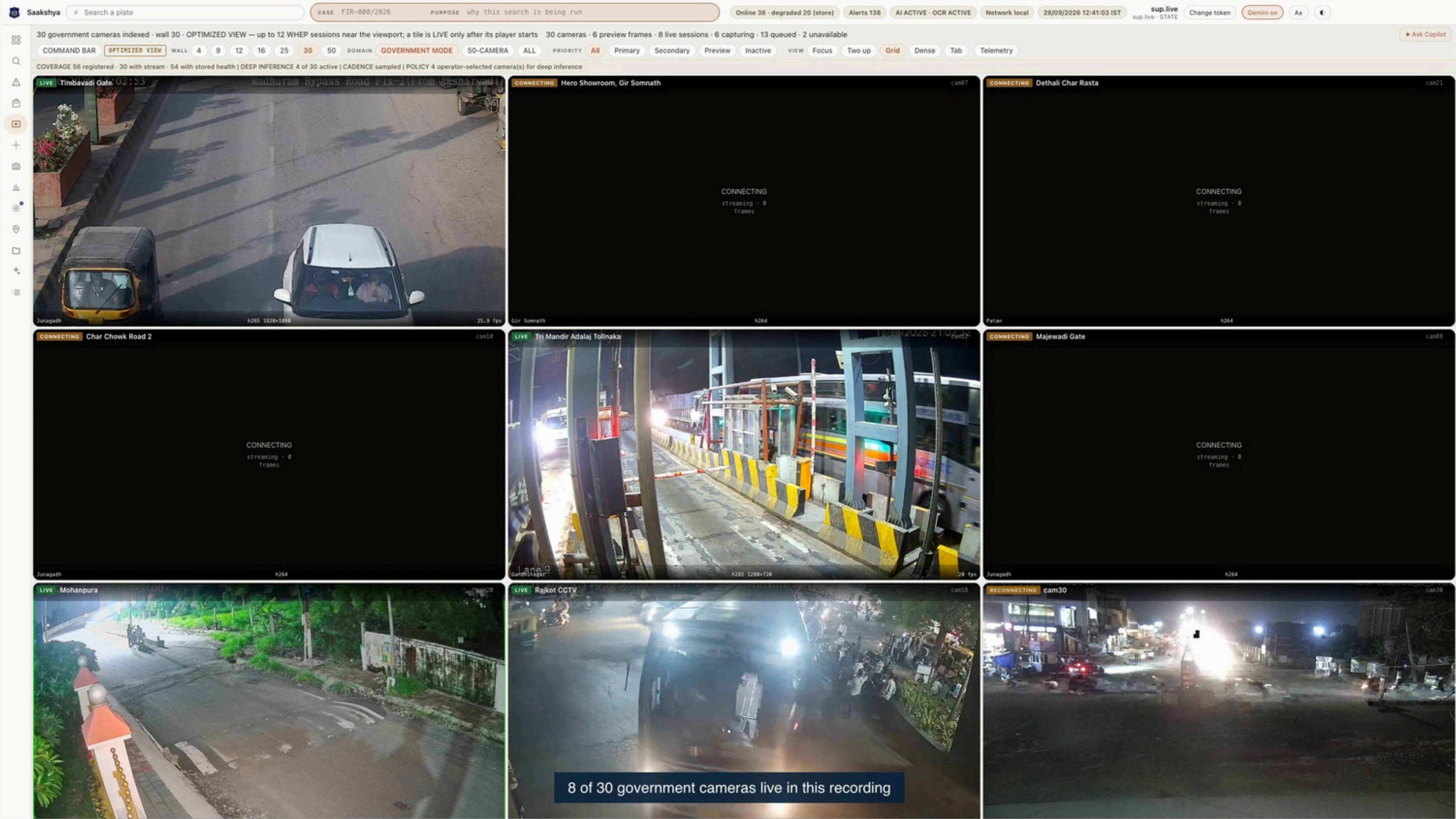
Playback progress governs LIVE; preflight gates and stall hand-over constrain recording. The ANPR export filters GOVERNMENT.

28 SEP · RECORDED GOVERNMENT DEMONSTRATION

Eight attempts (team-reported). MEASURED: 6–13 of 30 advancing; submitted take 5:40, 12:41–12:47 IST. 200 live cam06 reads in the 11:15–12:53 IST session; 21 within the filmed take. Single-camera evidence. 401 refusals from 12:57 IST prevented further recording. Cause unestablished; no fixed quota inferred.

Sources: docs/LIVE_INTEGRATION_STORY.md (claim-by-claim references); reports/PHASE14_REAL_CAMERA_SOURCE_CENSUS.md; reports/PHASE16_30_CAMERA_BROWSER_COVERAGE.md; docs/SENTINEL_SUPPORT_CLARIFICATION.md; reports/FINAL_SUBMISSION_CERTIFICATION.md; var/demo/gov_take7/beats.json.

Wall viewing and AI coverage are separate



Historical capture: 28 Sep 2026. CONTROL ROOM up to 30 WHEP sessions; OPTIMIZED VIEW ≤ 12 . Policies: ui/app.js. Live availability varies by test window; this filmed frame shows 8 live cameras. Source: var/demo/ui_shots_final/gov_live_grid.png

12 of 30 government cameras live in this recording

30 government cameras indexed · wall 30 · CONTROL ROOM — one WHEP session per tile, up to 30, staggered; a tile is LIVE only after its player starts 30 cameras · 1 preview frame · 17 live sessions · 6 capturing · 26 queued

COMMAND BAR

CONTROL ROOM

WALL 4 9 12 16 25 30 50

DOMAIN GOVERNMENT MODE

50-CAMERA

ALL

PRIORITY All

Primary

Secondary

Preview

Inactive

VIEW Focus

Two up

Grid

Dense

Tab

Telemetry

COVERAGE 56 registered · 30 with stream · 54 with stored health | DEEP INFERENCE 4 of 30 active | CADENCE sampled | POLICY 4 operator-selected camera(s) for deep inference

RECONNECTING can06

13-06-2026 18:00:34

Madhura Bypass Road P1x-2 (from deep inference)

Junagadh

CONNECTING can07

CONNECTING

streaming · 0 frames

Gir Somanth

CONNECTING can21

CONNECTING

streaming · 0 frames

Patan

CONNECTING can10

CONNECTING

streaming · 0 frames

Junagadh

LIVE can12

13-06-2026 21:00:10

Gandhinagar

CONNECTING can08

CONNECTING

streaming · 0 frames

Junagadh

LIVE can20

not on the map

LIVE can18

Rajkot

LIVE can30

not on the map

LIVE can01

13-06-2026 21:00:39

Chimara Bhangarage CSITMS PTZ2

Ahmedabad

LIVE can02

13-06-2026

Janpath T CSITMS-10_PTZ2

Gandhinagar

LIVE can03

13-06-2026

O.N.G.C. Office BS-103_B1

Ahmedabad

LIVE can04

Ahmedabad

LIVE can05

Ahmedabad

CONNECTING can09

CONNECTING

streaming · 0 frames

Junagadh

CONNECTING can11

could not capture a frame from this camera just now; the upstream feed may be unavailable or busy.

streaming · 0 frames

Junagadh

LIVE can13

Ahmedabad

LIVE can14

Ahmedabad

PREVIEW can15

13-06-2026 21:01:07

Ahmedabad

LIVE can16

2026-06-13 21:00:09

Visat T Junction P2 RLVD

Ahmedabad

LIVE can17

13-06-2026 17:00:17

1-CENTER

Rajkot

LIVE can19

Navsari

LIVE can22

14-06-2026 04:30

Banaskantha

LIVE can23

not on the map

CONNECTING can24

CONNECTING

streaming · 0 frames

not on the map

CONNECTING can25

READY

select for verified WHEP live video

not on the map

LIVE can26

13-06-2026 17:00:17

not on the map

CONNECTING can27

READY

select for verified WHEP live video

not on the map

CONNECTING can28

READY

select for verified WHEP live video

not on the map

CONNECTING can29

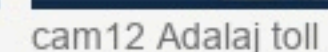
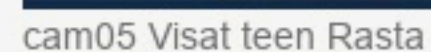
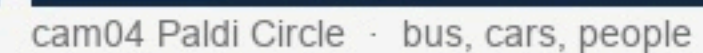
READY

select for verified WHEP live video

not on the map

12 of 30 government cameras live in this recording

Historical capture: 28 Sep 2026. 12 of 30 government cameras live in this recording. Measured in the 28 Sep 2026 film; availability varies with the test window. Source: var/demo/ui_shots_final/gov_live_dense_2026-09-28.png



Vehicle and person detection ran on the government feed

EVIDENCE	RESULT AND BOUNDARY
Observations · cam01–cam30	1,155,325 rows; 2 Sep 06:54 – 24 Sep 16:10 2026 IST. Counts are observations, not unique vehicles or people.
Vehicle detections	Car 529,966 (29 cameras); truck 184,600 (29); motorcycle 58,423 (25); bus 49,403 (28); bicycle 13,312 (25).
Person and other detections	Person 307,290 (all 30 cameras); truck_bus 5,819; van 5,049; unknown 1,463. Vehicle AND person detection are measured.
ANPR · 24 Sep snapshot	901 reads; 178 distinct plates; 97 confirmed registrations; 9 cameras. Confirmed = at least one read with ≥2 agreeing frames. Reads: 2–21 Sep 2026 IST.
Live session · 28 Sep 11:15–12:53 IST	4,465 observations on the 4 deep-inference cameras; 200 plate reads (124 plates) on cam06 during the recording session; 21 fall inside the 12:41–12:47 film. Delivered CSV: 1,101 reads, 264 plates, 9 cameras.
Government tracking finding	No plate was read on two government cameras: every government vehicle history is SINGLE-CAMERA. No appearance embeddings were stored; no government appearance re-identification ran.
Restricted-zone rule · DEMO	cam12: ‘No pedestrians on the toll-lane carriageway’. Person-class polygon rule set by the estate administrator for this evaluation; not a measured intrusion-detection accuracy result.

Sources: var/live.db observations and zone_rules (read-only SQL, 28 Sep 2026); var/demo/government_feed_anpr_report.csv; src/saakshya/reports/anpr.py::CONFIRM_VOTES.

Historical ANPR gallery · 24 Sep snapshot

901 total reads · 178 distinct plates · 97 confirmed registrations · 9 cameras

Read window: 2 Sep 2026 – 21 Sep 2026 (IST)

Selected examples below; statistics cover every government read in the stated window.



GJ11S7924
cam06 · OCR confidence 96.5%
6 agreeing reads
21 Sep 2026 · 02:13:45 IST



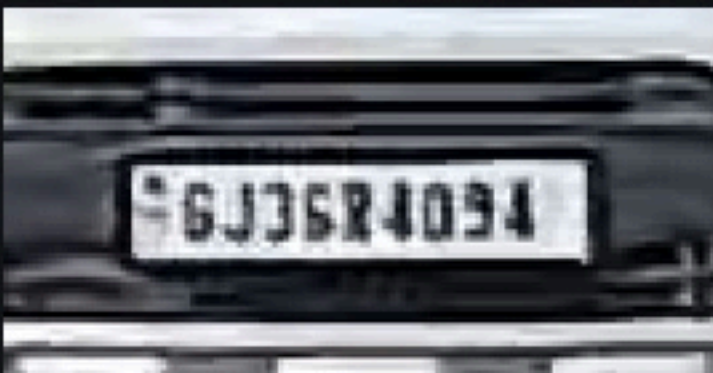
GJ04Z0610
cam06 · OCR confidence 93.2%
single read — a lead, verify
21 Sep 2026 · 06:29:54 IST



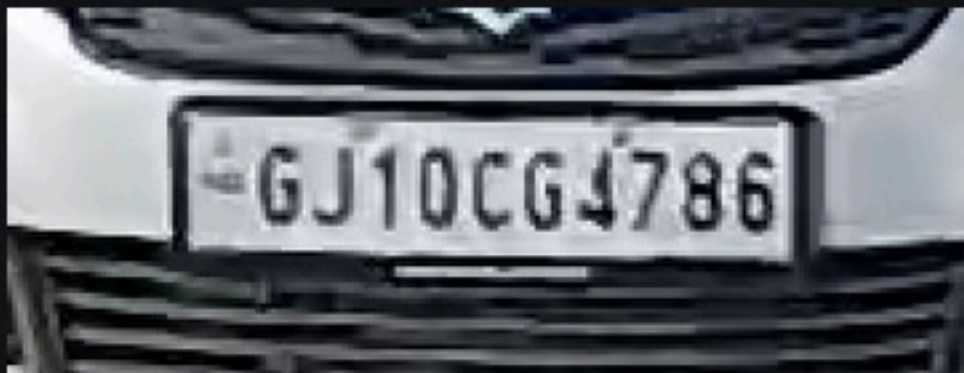
GJ11CH4401
cam06 · OCR confidence 92.4%
4 agreeing reads
21 Sep 2026 · 06:50:25 IST



GJ10CG4786
cam06 · OCR confidence 98.8%
single read — a lead, verify
21 Sep 2026 · 07:47:52 IST



GJ36R4094
cam06 · OCR confidence 99.0%
single read — a lead, verify
21 Sep 2026 · 07:47:53 IST



GJ10CG4786
cam06 · OCR confidence 94.0%
5 agreeing reads
21 Sep 2026 · 11:26:24 IST

Sources: var/demo/plate_gallery/selected.json (plate_image, nearest-neighbour enlargement of the sealed frame); stats.json (stated window only). var/live.db. Delivered CSV adds 200 reads from the 28 Sep session: 1,101 reads / 264 plates. Pre-28 Sep 11:15 IST stills may show another vehicle; hashes verify bytes only; check crop/read pairing.

Historical capture: 24 Sep 2026. Heads blurred from the person detector's boxes — no face detector exists. Live AI plane figures are measured, on the GPU. Source: `var/demo/ui_shots_final/own_intel.png`

Historical capture: 24 Sep 2026. SYNTHETIC RENDERED TEST CORPUS — route-logic demonstration, not camera footage. UI 'PLATE-CONFIRMED 2 of 2' counts plate-bearing route observations; 'TRAJECTORY CONFIRMED' is a route score. The trace report has 7 single-frame leads, 0 confirmed reads (var/demo.db; intelligence/trajectory.py). Source: var/demo/ui_shots_final/own_find.png

DEMO

Government stand-in and synthetic route logic

SOURCE	REGISTRATION / LOCATIONS	WHAT IT ESTABLISHES
Government stand-in	GJ11S7924 · 57 reads · cam06 only	Team-chosen from cam06 reads on 20 Sep, not organiser-issued. 52 snapshot + 5 session reads; all 5 precede the filmed take. No multi-camera route.
All government plate histories	1,101 reads · 264 distinct plates · 9 cameras	No plate appears on two government cameras. No appearance embeddings were stored; no appearance re-identification ran.
Synthetic route-logic report	GJ18JX7786 · C-014 → C-021	SYNTHETIC RENDERED TEST CORPUS — route-logic demonstration, not camera footage. Seven single-frame leads, zero confirmed reads. No real multi-camera evidence.
Older sealed stills	Before 28 Sep 2026 11:15 IST	May show another vehicle. A verified hash proves the bytes only; verify the crop/read pairing. These stills do not establish vehicle identity.

Enter the issued plate, then search the whole estate

STEP	OFFICER PROCEDURE
1 · Record the request	Enter the organiser-issued plate under a case and purpose. GJ11S7924 was a team-chosen rehearsal stand-in, not that issued number.
2 · Search retrospectively	Search the whole authorised estate and available history. Review timestamped camera locations and any supported trajectory legs; keep absent coverage explicit.
3 · Watch for new sightings	File the plate on the watchlist with its evaluation category, authority, scope and expiry so subsequent matching reads raise live alerts.
4 · Open the trace report	Review reads, camera locations, timestamps and evidence cautions, then export the trace report. A single-camera history is not a complete multi-camera route.

Sources: organiser FAQ Q27–28; docs/HLD.md §11; src/saakshya/reports/vehicle_trace.py. Procedure for evaluation day; no successful real multi-camera trace is claimed.

INVESTIGATE · LOOKALIKE, NOT A MATCH

6J1VV0119 is an OCR lookalike of GJ1VV0119

Saakshya

Gujarat Police · CCTV evidence

Search cameras, plates, cases

CASE

FIR-000/2026

PURPOSE

live government grid verificatio

Supervisor (live grid)

supervisor.live · STATE

Change token

Gemini off

A±

🔊

Official · sensitive

Law-enforcement use only. Every query is attributed and written to a hash-chained audit log.

live.db · LIVE CAPTURE

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REGISTRATION MARK

6J1VV0119

Allow near matches

Watchlist only

COLOUR

any

TYPE

car, person, bus

FROM

dd/mm/yyyy, --:-- --

TO

dd/mm/yyyy, --:-- --

DISTRICT

in jurisdiction

CAMERA

any

SEARCH

OBSERVATIONS 2 observations

One camera only

2 reads of this mark, all on cam07. That is one camera — looping footage or repeated passes — not a multi-camera route. Cross-camera identity is not claimed.

GJ1VV0119

REQUIRES VERIFICATION

2026-09-04 14:18:26

cam07 Hero Showroom, Gir Somnath

UNCORROBORATED LEAD

quality 0.57

GJ1VV0119

REQUIRES VERIFICATION

2026-09-04 14:18:26

cam07 Hero Showroom, Gir Somnath car black

quality 0.57

EVIDENCE

MOVEMENT 6J1VV0119

Fit

Attach to case

COLOUR BY

Health

ANPR capability

Analytics tier

LAYERS

Alerts

Coverage gaps

Labels

LEGEND

TRAJECTORY

LIKELY · score 0.69 · 1 camera

cam07

14:18:28

quality 0.57

Timebase · RESTRICTED

A single camera. There is no interval between sightings to reason about, so no shared timebase is needed or claimed. This camera's own timing has not been established, so even its position on a timeline is provisional.

STATUS

LIKELY

SCORE

0.69 — ordering score, not a probability

DURATION

0 s

PLATE-CONFIRMED

1 of 1

MEAN QUALITY

0.57

2 track fragments of the same registration mark at cam07 within 120s — one pass, not several. Kept the highest-quality view.

WHY THIS MATCH

0B01M1PCJ1E0TRMRXE4FF2MC06NP

WHY THIS MATCH

REQUIRES VERIFICATION

PLATE

0.70 × 0.45

Weakest signal

plate: stored mark GJ1VV0119 is one OCR confusion from queried 6J1VV0119 (one-character repair 6->G at position 0); the stored read was not edited

Score is an ordering score over candidates, not a calibrated probability.

matched via a known OCR confusion (O/O, B/8, ...), not an exact registration mark — verify before acting

SOURCE

CAMERA

cam07 · Hero Showroom, Gir Somnath

DISTRICT

Gir Somnath

DEPARTMENT

—

LOCATION

20.91370, 70.36730

PTS

79.338 s

NORMALISED

2026-09-04 14:18:26

TRACK

TR01M1PCJ1DRVQNV5NCC3NBAG344

TYPE

unknown

PLATE VOTES

1 frame — uncorroborated lead

QUALITY

0.57 (grade UNKNOWN)

SHARPNESS

1.0

LUMINANCE

0.4

PLATE WIDTH

83 px

OCR RAW

GJ1VV0119

MODELS

model=anpr-onnx-cpu@1.0.0 task=plate_detect_ocr hub_id=yolo-v9-t-640-license-plate-end2end revision=null runtime=fast-alpr precision=fp32 backend=fast-alpr/onnx

EVIDENCE

Not yet sealed

No evidence record has been created for this observation. Sealing computes digests of the frame and clip and appends a hash-chained manifest.

SEAL EVIDENCE

ATTACH

30 cameras onboarded

0 graded good for plate reading

75 marks read in the last hour

1 alerts open, unacknowledged

Searched 30 camera(s) in STATE · 30 excluded: CAMERA_UNAVAILABLE

Historical capture: 15 Sep 2026. Historical lookalike example, separate from the designated vehicle. The platform does not infer another camera or a route from OCR similarity. Source: `var/demo/ui_shots_final/gov_find_lookalike.jpg`

MEASURED

INVESTIGATE

35

Historical capture: 24 Sep 2026. Shown entries are evaluation marks. Pre-28 Sep 11:15 IST stills may show another vehicle; hashes verify bytes only. The store also has a representative stolen_vehicle entry (var/live.db). Source: var/demo/ui_shots_final/gov_alerts_film.jpg

MEASURED

Historical capture: 24 Sep 2026. SYNTHETIC RENDERED TEST CORPUS — route-logic demonstration, not camera footage. Fictional plates and listings demonstrate incident grouping, not a government theft record (var/demo.db; tools/sandbox/make_media.py). Source: var/demo/ui_shots_final/own_alerts.png

DEMO

Historical capture: 24 Sep 2026. SYNTHETIC RENDERED TEST CORPUS — route-logic demonstration, not camera footage. Fictional plate/listing; the displayed police attribution is part of the demonstration. Seven single-frame leads, zero confirmed reads (var/demo.db). Source: var/demo/ui_shots_final/own_trace_report.png

DEMO

What each sealed record is, and the link that binds it



Saakshya

Gujarat Police · CCTV evidence

CASE FIR-214/2026

PURPOSE tracing a designated vehicle

Online 6 · degraded 49 (store)

Alerts 2

AI ACTIVE · OCR ACTIVE

Network local

24/09/2026 11:18:08 IST

Insp. R. Desai
sup.fim · STATE

Change token

Gemini on

As

Official · sensitive Law-enforcement use only. Every query is attributed and written to a hash-chained audit log.

film.db · LIVE CAPTURE Deployment details

OPERATIONS INTELLIGENCE **INVESTIGATION** SYSTEM

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Investigate

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Intelligence

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Estate map

Cases 1

Copilot

Audit log

Evidence chain verified

Every manifest hashes to the value recorded in the entry that follows it. A removed or altered record would break this.

#	SEALED RECORD	CAMERA	CAPTURED (IST)	PLATE	STILL	CHAINED TO PREVIOUS	CERTIFICATE	VERIFIED
1	EZ01M2TW9EB5D3H5EA6810FDEQ4B	Naroda Circle Junction C-014	01 Sep 2026 13:30:22 IST	GJ18JX7786	no still	genesis → 565fa553...	draft pending signature	✓ verified
2	EZ01M2TW9EB6Z9P4K3J99JRPVX8Z	Naroda Circle Junction C-014	01 Sep 2026 13:30:23 IST	GJ18JX7786	no still	565fa553... → 1578cdc2...	draft pending signature	✓ verified
3	EZ01M2TW9EB7QBWYFVVT77MY2CKB	GSRTC Depot Gate 2 C-021	01 Sep 2026 13:35:20 IST	GJ18JX7786	no still	1578cdc2... → fd2996e2...	draft pending signature	✓ verified
4	EZ01M2TW9EB7B88KB1QKF061ZPHA	GSRTC Depot Gate 2 C-021	01 Sep 2026 13:35:21 IST	GJ18JX7786	no still	fd2996e2... → 3f266a3f...	draft pending signature	✓ verified
5	EZ01M2TW9EB84P35R698Q0G39913	GSRTC Depot Gate 2 C-021	01 Sep 2026 13:35:22 IST	GJ18JX7786	no still	3f266a3f... → 101b05fb...	draft pending signature	✓ verified
6	EZ01M2TW9EB9J9QRCCQM017XDG2	GSRTC Depot Gate 2 C-021	01 Sep 2026 13:35:23 IST	GJ18JX7786	no still	101b05fb... → 004cc82a...	draft pending signature	✓ verified

54 cameras onboarded

5 graded good for plate reading

126 marks read in the latest hour

2 alerts open, unacknowledged

Historical capture: 24 Sep 2026. SYNTHETIC RENDERED TEST CORPUS — route-logic demonstration, not camera footage. Records: var/demo.db. Hashes establish file integrity only. Pre-28 Sep 11:15 IST stills may show another vehicle; verify crop/read pairing. Source: var/demo/ui_shots_final/own_evidence.png

DEMO

EVIDENCE 39

Historical capture: 28 Sep 2026. Grades are computed from each camera's own stream. UNKNOWN means there is not yet enough evidence to grade. Source: [var/demo/ui_shots_final/gov_cameras_film.jpg](#)

MEASURED

Historical capture: 28 Sep 2026. Markers show camera health. Missing coordinates stay in the registry strip; positions derived from camera names retain their provenance. Source: [var/demo/ui_shots_final/gov_map_film.jpg](#)



It will not enhance a still — that would be fabricated evidence



Saakshya
Gujarat Police · CCTV evidence

Search cameras, plates, cases

CASE

FIR-000/2026

PURPOSE

live government grid verificati

Supervisor (live grid)

supervisor.live · STATE

Change token

Gemini off

A±

Official · sensitive

Law-enforcement use only. Every query is attributed and written to a hash-chained audit log.

live.db · LIVE CAPTURE

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INVESTIGATOR COPILOT 16 read-only tools · rules

One coordinator over four deterministic specialists. Stills stay on this host unless vision is switched on for a supervised demo.

ROUTES THE QUESTION TO SPECIALISTS. NARRATES. INVENTS NOTHING.

REGISTRY, CAPABILITY, INFRARED, LOCATION BASIS. NEVER INVENTS COORDINATES.

PLATE SEARCH, WATCHLIST, MATCH EXPLANATION. LEADS STAY LEADS.

WHETHER TWO CAMERAS MAY SHARE A TIMELINE. DEFAULT IS RESTRICT, NOT JOIN.

SEALED MANIFESTS AND VERIFICATION. REFUSES IMAGE ENHANCEMENT.

Infrared cameras

Cannot read plates

Unlocated

Timebase cam01+cam21

Timebase cam01+cam04

Find GJ32AG0028

Enhance a still

search_plate: result was truncated build_trajectory: 1 hypothesis(es); the best is LIKELY with score 0.689 across 0 leg(s). A score orders candidates and is not a probability. Answered from the tool results below by deterministic rules; no language model was involved, and no data left this deployment.

Specialists: Identity · timebase



cam07

ingest still · not evidence

Describe still



cam01

ingest still · not evidence

Describe still



cam02

ingest still · not evidence

Describe still



cam03

ingest still · not evidence

Describe still



cam04

ingest still · not evidence

Describe still



cam05

ingest still · not evidence

Describe still



cam06

ingest still · not evidence

Describe still



cam08

ingest still · not evidence

Describe still



cam09

ingest still · not evidence

Describe still



cam10

ingest still · not evidence

Describe still



cam11

ingest still · not evidence

Describe still



cam12

ingest still · not evidence

Describe still



cam13

ingest still · not evidence

Describe still



cam14

ingest still · not evidence

Describe still



cam15

ingest still · not evidence

Describe still



cam16

ingest still · not evidence

Describe still



cam17

ingest still · not evidence

Describe still



cam18

ingest still · not evidence

Describe still



cam19

ingest still · not evidence

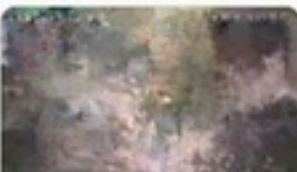
Describe still



cam20

ingest still · not evidence

Describe still



cam21

ingest still · not evidence

Describe still



cam22

ingest still · not evidence

Describe still



cam23

ingest still · not evidence

Describe still



cam24

ingest still · not evidence

Describe still

Queries run: search_plate build_trajectory

Enhance this still and sharpen the plate so I can read it.

refuse_imagery: refused — Government stills are not enhanced, sharpened, generated or inpainted. A plate invented or 'restored' by a model would be fabricated evidence (BSA s.63). Detection and ANPR stay on this host. Ask list_estate to show the real cameras instead. Answered from the tool results below by deterministic rules; no language model was involved, and no data left this deployment.

Specialists: evidence

Queries run: refuse_imagery (refused)

e.g. show the infrared cameras, or may cam01 and cam21 share a timeline?

30cameras onboarded

0graded good for plate reading

75marks read in the last hour

1alerts open, unacknowledged

Historical capture: 15 Sep 2026. refuse_imagery. Detection and ANPR stay on this host. No language model is allowed to invent a plate. Source: var/demo/ui_shots_final/gov_copilot_refuse.jpg

MEASURED

COPILOT 42

cam01 and cam21 may not share a timeline



Saakshya
Gujarat Police · CCTV evidence

🔍 Search cameras, plates, cases

CASE **FIR-000/2026**

PURPOSE **live government grid verificati**

Supervisor (live grid)
supervisor.live · STATE

Change token

Gemini off

A±

🔊

Official · sensitive Law-enforcement use only. Every query is attributed and written to a hash-chained audit log. live.db · LIVE CAPTURE

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One coordinator over four deterministic specialists. Stills stay on this host unless vision is switched on for a supervised demo.

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PLATE SEARCH, WATCHLIST, MATCH EXPLANATION. LEADS STAY LEADS.

WHETHER TWO CAMERAS MAY SHARE A TIMELINE. DEFAULT IS RESTRICT, NOT JOIN.

SEALED MANIFESTS AND VERIFICATION. REFUSES IMAGE ENHANCEMENT.

Infrared cameras

Cannot read plates

Unlocated

Timebase cam01+cam21

Timebase cam01+cam04

Find GJ32AG0028

Enhance a still

cam11

ingest still · not evidence

Describe still

cam12

ingest still · not evidence

Describe still

cam13

ingest still · not evidence

Describe still

cam14

ingest still · not evidence

Describe still

cam15

ingest still · not evidence

Describe still

cam16

ingest still · not evidence

Describe still

cam17

ingest still · not evidence

Describe still

cam18

ingest still · not evidence

Describe still

cam19

ingest still · not evidence

Describe still

cam20

ingest still · not evidence

Describe still

cam21

ingest still · not evidence

Describe still

cam22

ingest still · not evidence

Describe still

cam23

ingest still · not evidence

Describe still

cam24

ingest still · not evidence

Describe still

Queries run: search_plate build_trajectory

Enhance this still and sharpen the plate so I can read it.

refuse_imagery: refused — Government stills are not enhanced, sharpened, generated or inpainted. A plate invented or 'restored' by a model would be fabricated evidence (BSA s.63). Detection and ANPR stay on this host. Ask list_estate to show the real cameras instead. Answered from the tool results below by deterministic rules; no language model was involved, and no data left this deployment.

Specialists: evidence

Queries run: refuse_imagery (refused)

May cam01 and cam21 share a timeline?

check_timebase: cam01 and cam21 → REFUSED. cam21 has UNRELIABLE timing (140 PTS regressions) — its observations cannot be placed on any timeline Answered from the tool results below by deterministic rules; no language model was involved, and no data left this deployment.

Specialists: timebase

cam01

ingest still · not evidence

Describe still

cam21

ingest still · not evidence

Describe still

Queries run: check_timebase · 3 facts checked against results

e.g. show the infrared cameras, or may cam01 and cam21 share a timeline?

30 cameras onboarded

0 graded good for plate reading

75 marks read in the last hour

1 alerts open, unacknowledged

Historical capture: 15 Sep 2026. REFUSED: cam21 has unreliable timing (PTS regression). Clocks are not joined by narrative. Source: var/demo/ui_shots_final/gov_copilot_timebase.jpg

MEASURED

COPILOT 43

Decode capacity and deep inference have different limits

RUN / SOURCE	MEASURED RESULT	BOUNDARY
camera_load.json · 2026-09-01	50 requested; 44 streaming and 6 down at end	Controlled load run; this is not concurrent government deep inference.
camera_load.json · decoder	0 decoder errors; 36 open failures	Failed streams did not all recover by the end of the run.
camera_load.json · analysis	11.4 fps per process; 4,903.3 MB peak RSS	This workload is separate from the device benchmark and the HLD sizing baseline.
live_cluster.json · 2026-09-02	6 camera cohort; 6,242 frames analysed / 1,072.6 s	Historical government test window. Decoded fps is not deep-inference fps.

Sources: var/reports/camera_load.json; var/reports/live_cluster.json. Simultaneously live camera counts are measured during a test window, never a sandbox or bridge limit.

Measured device throughput, with the sample boundary

WHAT WAS TIMED	CPU	APPLE GPU (MPS)	SPEED-UP	SAME OUTPUT?
Detector alone, 2560×1440 frame	448 ms	133 ms	3.36×	3,622 of 3,622 boxes at IoU ≥ 0.9
Whole pipeline: detect, track, plate, OCR	598 ms	293 ms	2.04×	same observations (no plates in sample)

Sources: var/reports/detector_device.json; pipeline_device.json. Historical integrated-GPU benchmarks; earlier OCR pipeline with plate models on CPU. Neither a target-accelerator result nor OCR accuracy parity.

AI coverage and hardware scaling

Every integrated camera viewable on the wall, subject to availability; 4 deep-inference slots by default, prioritised by measured capability; the live worker samples at a fixed interval.

VERIFIED default: analytics/worker.py::AI_CAMERA_LIMIT; assigned at boot, not rotated. Historical cohort: 6 cameras analysed in live_cluster.json; not current concurrency.

CURRENT DEMO MACHINE

MEASURED

Historical live_cluster.json cohort: 6 cameras with analysed frames. This is not the current slot count or a hardware maximum.

Device pipeline: 3.42 fps; 292.8 ms median / 374.7 ms p95. pipeline_device.json.

Detector only: 7.5 fps; 133.3 ms median. detector_device.json.

Decode/load: 44/50 streaming at end. camera_load.json. Separate workload.

Earlier OCR pipeline; current recogniser needs a fresh benchmark. Slots are assigned GOOD > DEGRADED > UNKNOWN > UNSUITABLE at worker boot.

CELL GPU POOL

MODELLED

SIZED: 2,500 cameras per full cell at 1 Hz, ANPR-grade 20% at 5 fps (ASSUMED); 5.6 fps CPU sizing baseline. S=10 speedup ASSUMED, not measured on target GPUs.

83 GPUs / full cell in 21 four-GPU servers. 80,000 cameras in 40 cells → 2,652 GPUs / 663 servers (1 Hz for all: 1,509 / 378).

Decode is sized separately. Benchmark tendered hardware, including OCR and memory, before procurement.

docs/HLD.md §17.6, §20.2–20.3. Planning input differs from the device workload at left.

STATEWIDE

DESIGNED

Regional ingest; horizontal worker pools; metadata event bus and district-local queues.

AdaptiveInferenceScheduler (built; tested in the certification harness) sets cadence by priority mode; live-worker wiring is DESIGNED.

Camera assignment and failover across a worker cluster require deployment validation. No camera-rotation claim.

The 80,000 figure is a registry/GIS metadata load test with synthetic camera rows plus architecture sizing.

Sources: var/reports/{live_cluster,pipeline_device,detector_device,camera_load}.json; docs/HLD.md §20; src/saakshya/{analytics/worker,runtime/inference_scheduler}.py.

A proposal that lists only strengths cannot be checked

- Prototype scope: operational deployment, evidentiary acceptance and statewide live-video scale still require validation.
- ANPR yield depends on camera geometry and light. A camera being viewable does not mean it is under deep inference.
- GJ11S7924 on cam06 is a SINGLE-CAMERA GOVERNMENT OBSERVATION; no government route is established.
- GJ18JX7786 on C-014 → C-021: SYNTHETIC RENDERED TEST CORPUS — route-logic demonstration, not camera footage.
- Person boxes are presence. There is no face identification on government data.
- Own-feed footage is published with heads blurred from the person detector's boxes. A person it never found is not blurred.
- The GPU speed-up is measured on a laptop's integrated GPU. The target accelerator's is not quoted until it is run.
- Copilot is read-only. It will not enhance a still, invent a plate, or join clocks the timebase refuses.
- No government plate spans two cameras; no government appearance embeddings were stored. Pre-28 Sep 11:15 IST stills may show another vehicle; hashes verify bytes only.
- The government store includes an active representative stolen_vehicle entry, GJ07XZ4409, with no government reads; its listing is not a verified theft record.

Three films. One plate report. No mock-ups.

own_feed.mp4 · controlled own feed 2:43 · 2560×1440 · audio present

Licensed Mumbai street footage, heads blurred. Masked sign-in; a camera onboarded through the registry form, validated before write; the administrator → officer handoff; the production pipeline's per-frame boxes replayed on the video clock, plates drawn only when the vote holds; a plate searched; a fictional watchlist alert, its route and trace report (synthetic corpus, labelled); the evidence chain.

government_feed.mp4 + ANPR report CSV 5:40 · 2560×1440 · audio present

The government grid, recorded live 28 Sep 12:41–12:47 IST: onboarded registry (no onboarding action), 30 control-room tiles with varying live availability, a focused camera, person detections and the demonstration zone rule, the ANPR gallery, the single-camera trace, GIS and the trace report. The CSV holds every government read: 901 to the 24 Sep snapshot and 200 during the 28 Sep recording session (11:15–12:53 IST). Only 21 reads fall inside the filmed take.

government_tour.mp4 · every feature, from sign-in 15:04 · 2560×1440 · audio present

RECORDED GOVERNMENT FOOTAGE (captured 15 Sep, replayed, never called live) on the wall and in focus, with the pipeline's per-frame boxes; then alerts, searches, evidence verification, cases and export, the Gemini copilot, the audit chain, Model 1 grades, gaps and validation, Model 3 systems, Model 4 analytics, the administrator refusal and the handoff.

Sources: var/demo/*.mp4 (ffprobe at render). Own-feed plates come from the final pipeline's sidecars (Indian-trained recogniser); government session reads are the current recogniser's (docs/FINAL_SUBMISSION.md).

Evaluation framework, mapped

SEVEN COMMON AREAS

01 Government test case

Government film + ANPR CSV: camera and timestamp; no proven multi-camera government route.

02 Presentation clarity / completeness

This deck: model, workflow, scenario, sizing, evidence and explicit limits.

03 Architecture / feasibility / security

HLD diagrams + §10 adapters, §17 sizing, §18 security; SECURITY.md.

04 Working platform maturity

Own film: form onboarding and admin → officer handoff. Government film: onboarded registry, viewing and analytics; no onboarding action.

05 Analytics output quality

Person/vehicle detection; cam12 restricted-zone entries under an administrator DEMONSTRATION rule in the film. Intrusion accuracy unmeasured.

06 ~80k scale / PoC readiness

05_SCALE_80K_LOAD_TEST.md + HLD §17: registry load evidence; statewide live AI remains modelled.

07 Accessible, consistent submission

00_SUBMISSION_INDEX.md indexes docs, films and reports; public links/access need final verification.

SIX BONUS AREAS

Hybrid architecture

Models 1 + 2 + 3 and selected-camera M4 analytics; HLD architecture diagrams.

Cross-camera vehicle tracking

C-014 → C-021: SYNTHETIC RENDERED TEST CORPUS — route-logic demonstration, not camera footage.

Reliable analytics beyond ANPR

Government detection overlays, person/vehicle boxes, tracking and measured capability grades; no face identity claim.

Edge / bandwidth / low connectivity

HLD §6, §12 + SCALE_MODEL: durable queue, offline replay, metadata aggregation.

Cybersecurity / privacy / audit / RBAC

SECURITY.md + Evidence and Audit views: hash chains, jurisdiction gates, blurred heads in own film.

Dashboards / alerts / health / APIs

Overview, Alerts and Cameras views; 05_* registry API and gap-analysis material; HLD §10 adapters.

Evidence index, not a score claim. Film/report descriptions follow docs/FINAL_SUBMISSION.md; delivery links are verified at submission.



Find → Trace → Verify → Act.

Analytics at the edge. Metadata to the centre.

Video stays where it is.